

Article

Using Digital Twin Technology to Improve the Organization of the Supply Chain in Piece Type of Production

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Abstract

Digital twin technology has proven to be a transformative enabler for sustainable manufacturing by providing real-time virtual representations of physical assets and supply chain processes. This paper explores the integration of digital twins with agile supply chain strategies to improve the sustainability of manufacturing systems. By leveraging real-time data and advanced simulations, digital twins facilitate dynamic decision making, optimize resource utilization and reduce environmental impact. A case study is presented in which a digital twin is implemented with the aim of improving the responsiveness of agile supply chains and suggesting appropriate times for the delivery of components and the shipment of the final product, with the goal of minimizing the time components spend in warehouses. The analysis shows how digital twins improve clarity, adaptability and predictive capabilities, leading to greater efficiency and sustainability. The results show that the combination of digital twin technology and agile supply chain frameworks contributes significantly to resource optimization, emissions reduction and overall operational resilience. The proposed approach proves to be highly effective for various manufacturing environments, especially those that strive to balance efficiency and sustainability goals.

Keywords: digital twin; digital environment; supply chain; sustainability; simulation; methodology



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1. Introduction

The emergence of Industry 4.0 has profoundly changed the manufacturing landscape and promoted the development of smart, connected and highly automated production environments. Modern manufacturing systems are rapidly evolving to meet growing demands for agility, flexibility and sustainability while ensuring high product quality and adapting to changing market dynamics [1–5]. As the industry transitions to data-driven operations, the integration of advanced technologies is no longer optional, but essential to maintaining competitiveness.

A key enabler of this transformation is digital twin technology, which enables the creation of dynamic virtual replicas of physical systems. These digital models are continuously synchronized with real-time data, enabling manufacturers to gain actionable insights, perform predictive analytics and optimize operational performance.

To effectively show its potential, a digital twin must represent the real system in sufficient detail to capture the most important behaviors and parameters required for monitoring, analysis and optimization [6,7]. These models simulate production operations, identify potential problems and evaluate performance in a risk-free virtual environment.

The origins of digital twin technology can be traced back to the early 2000s, with its first formal definition introduced in 2003 [8,9]. Since then, numerous researchers have refined the concept and explored its applications in manufacturing, engineering and systems management [10–13]. A widely accepted definition describes the digital twin as a highly detailed simulation model that integrates real-time data and physics-based algorithms to replicate the behavior and life cycle of a physical system [14,15].

As sustainability becomes a key priority in industrial development, more and more studies are exploring how digital twins can support sustainable practices [16–21]. In addition to traditional operational benefits—such as process optimization, improved quality and reduced unplanned downtime—digital twins are increasingly recognized for their potential to simultaneously improve environmental and economic performance. Sustainable manufacturing aims to balance economic growth and environmental responsibility by minimizing energy consumption, reducing greenhouse gas emissions and maximizing the reuse of materials throughout the product life cycle [22–25].

A key component of these goals is the integration of circular economy principles that promote the design of products and processes that facilitate recycling, reuse and resource efficiency. Embedding these principles within the framework of the digital twin enables manufacturers to align their production goals with both environmental and economic sustainability. This also contributes to cost savings through reduced material waste and more effective resource management.

In parallel, sustainable supply chains are becoming increasingly important to achieve both short-term efficiency and long-term competitiveness. These supply chains include practices such as ethical sourcing, environmentally friendly logistics and waste minimization [26,27]. Digital twins improve the sustainability and cost efficiency of supply chains by providing real-time visibility of material flows, predicting potential disruptions and optimizing logistics and inventory systems [28,29].

By simulating different supply chain scenarios, digital twins facilitate the introduction of greener and more cost-efficient logistics strategies—reducing overproduction, cutting emissions and optimizing transport routes and delivery schedules.

Despite the increasing importance of digital twin technology for supply chain optimization, there is still a lack of research addressing the practicality, transparency and cost-effectiveness of scenario-based decisions. Existing models often neglect the comprehensive testing of alternative transportation options for incoming components and finished products as well as the evaluation of different storage capacities. This approach has also attracted interest in real production environments, especially in smaller companies. In addition, there are few studies that integrate a total cost tracking—including the costs of transportation and rented storage capacity—while also incorporating sustainability aspects. This gap limits the wider application of digital twin solutions for operational decision support, especially for companies seeking accessible and customizable tools.

In supply chain management, decision-makers are under increasing pressure to optimize logistics processes and find a balance between cost efficiency and sustainability. However, traditional tools often lack the flexibility to test “what-if” scenarios in a transparent and traceable way. There is a need for a digital twin solution that is easy to use, cost-efficient and can simulate different transportation configurations and storage capacity scenarios. In addition, existing tools often do not provide a holistic view of the combined costs of transportation and warehouse rental and rarely integrate sustainability-oriented metrics into the decision-making process.

This article presents a novel, user-friendly digital twin framework designed to increase transparency and support the analysis of “what-if” scenarios in supply chain logistics. The proposed solution enables cost-effective simulation of alternative transportation and

storage capacity configurations, allowing decision makers to evaluate logistics strategies without disrupting actual operations. In addition, the model tracks total logistics costs—including transportation and warehouse rent—and incorporates sustainability aspects to support more responsible and data-driven decisions. The approach aims to bridge the gap between the theoretical potential and practical applicability of digital twins in supply chain management.

The rest of the paper is organized as follows. The Literature Review section provides an overview of existing contributions in the areas of digital models, digital twins, algorithms and various aspects of sustainable supply chains in manufacturing systems. The Materials and Methods section describes the methodology used to develop the digital twin, as well as a description of the experimental setup used for the analysis. Section 4, entitled Digital twin for testing system, demonstrates the functionality of the digital twin, including a detailed explanation of the algorithms that support its functionality. This is followed by the Results section, which presents the analyzed data using tables, graphs and charts. The findings are further interpreted in the Discussion section. The paper concludes with the Conclusions and Future Work section, which summarizes the key findings and outlines directions for future research.

2. Literature Review

This section provides a comprehensive overview of existing research on the application of digital twin technology in manufacturing and supply chain systems.

For a better understanding of the overall content, the difference between a traditional simulation model [30] and a digital twin [31] must first be explained. A traditional simulation model is a computational representation based on static or predefined input data. Although effective for analysing specific scenarios, it generally lacks real-time connectivity and interaction with physical systems, making it less capable of capturing dynamic, real-world behaviour. A digital twin is a virtual representation of a physical object, system or process that is continuously updated with real-time data from sensors, IoT devices or other sources. This dynamic model mirrors its real-world counterpart and enables continuous monitoring, analysis, prediction and optimization.

Traditional supply chains often struggle with limited transparency, making it difficult to track goods and materials throughout the system. Digital twin technology has proven to be a transformative solution for supply chain management [32,33], as it increases the efficiency of logistics and enables the simulation of different scenarios without disrupting the actual operational system [34–38].

A key advantage of digital twins for supply chain management is their ability to provide end-to-end transparency in real time. Digital twins aggregate data from IoT devices, ERP systems and sensors to create a constantly updated virtual model of the supply chain [39–43]. This improved visibility enables stakeholders to monitor product movements, identify bottlenecks and improve traceability. According to the authors Zhang et al. [44], digital twins support more effective inventory management by reducing the number of stock-outs and overstocks, thus improving the overall supply chain flow. This capability also increases responsiveness to fluctuations in customer demand, supply disruptions and logistical challenges. The authors also come to very similar conclusions in the study [45], in which digital twins can continuously monitor threats such as supply delays or demand shifts so that companies can take proactive measures and reduce vulnerability. They also support the modelling of alternative supply chain configurations, facilitate contingency planning and improve the ability to respond to sudden changes.

Digital twins are playing an increasingly important role in the context of sustainability [46,47]. By simulating multiple supply chain scenarios, companies can assess the

environmental impact of their operations and make data-driven decisions to promote greener practices. As highlighted in [48], digital twins facilitate the modeling of material, energy and resource flows, making it easier to identify inefficiencies and uncover opportunities to reduce waste. For example, manufacturers can use digital twins to refine production scheduling and inventory control to reduce excess material and waste. By modeling entire logistics networks, digital twins enable route optimization, greater fuel efficiency and improved delivery performance [49–52]. The research presented in [53] shows how digital twins are being used to streamline last-mile delivery and manage transportation fleets more effectively, reducing both operational costs and environmental impact. Their real-time capabilities also support dynamic responses to disruptions such as traffic congestion or delivery delays, enabling real-time adjustments to routes and schedules as conditions change.

Advanced algorithms and, increasingly, artificial intelligence play a crucial role in the effective use of digital twins in production management and throughout the supply chain [54–57]. Genetic algorithms are often used to tackle complex challenges such as production scheduling and resource allocation by efficiently exploring large solution spaces. Similarly, particle swarm optimization based on social behavior dynamics is used to simulate and optimize logistics and supply chain networks in real time. Reinforcement learning further enhances digital twins by enabling them to adaptively learn optimal strategies through continuous interaction with their environment, especially in dynamic decision-making scenarios. In addition, machine learning techniques, which include both supervised and unsupervised learning, are widely used for tasks such as predictive maintenance, demand forecasting and anomaly detection.

3. Materials and Methods

This paper presents a comprehensive methodology for improving the scheduling and coordination of component deliveries and the shipment of final products in a production-oriented company. The methodology aims to optimize logistical processes by selecting suitable transport vehicles for both the delivery of raw materials and the shipment of finished products. It also supports the precise scheduling of delivery and shipping dates in coordination with incoming customer orders and production planning.

A set of special algorithms have been developed to facilitate the implementation of this methodology. These include part control algorithms that ensure accurate tracking and verification of individual components, part counting that automates the quantification of components, and material inventory detection that enables real-time monitoring and evaluation of stock levels. Together, these algorithms form a robust framework that improves the efficiency, accuracy and responsiveness of the company's logistics and supply chain operations (Figure 1).

The basis of the entire methodology presented in this paper is the digital twin of a real manufacturing system. The digital twin was developed using a five-step methodology developed in our laboratory, which has been described in detail in previous studies [58–60].

For a better understanding, the following section provides a detailed overview of the manufacturing system for which the supply chain was developed. The real pallet-based manufacturing system consists of three assembly stations, each of which is followed by quality control operation. The assembly process takes place on three pallets on which two different product types—Type_N and Type_B—are produced (Figure 2). The main difference between these two product types is the width of the fabric and metal sheets used for production.

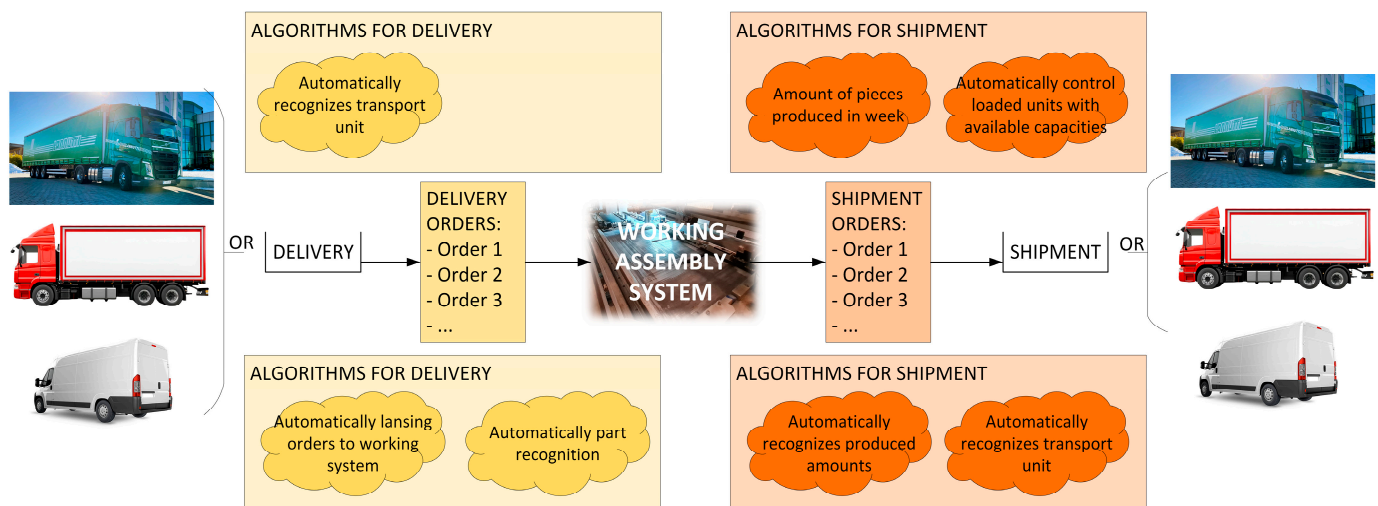


Figure 1. Logical schematic representation of a digital twin for component delivery in the production process and the final shipment of finished products.

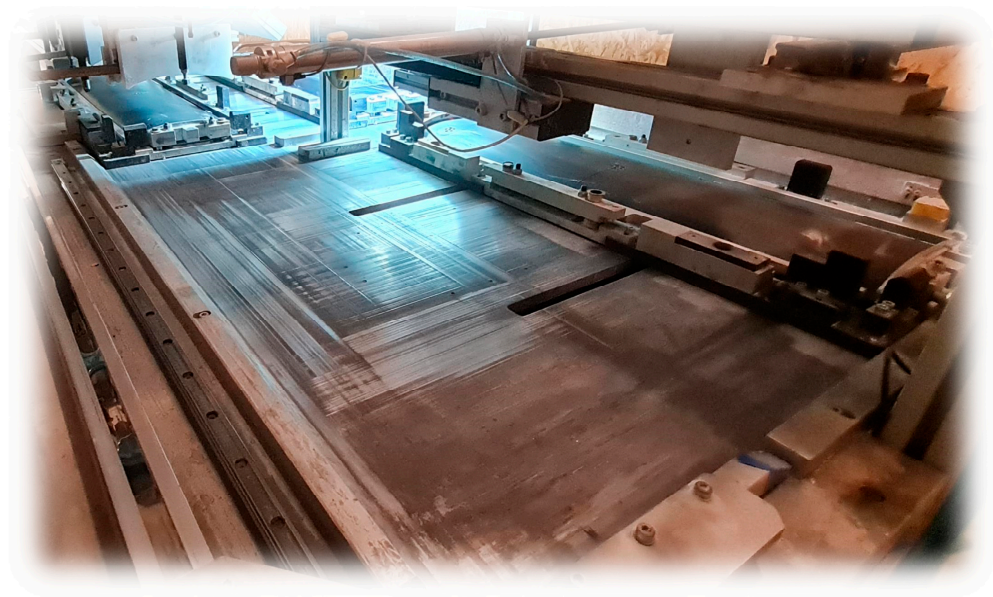


Figure 2. Real manufacturing system for the tests.

Although the overall manufacturing process is similar for both product types, specific adjustments to the production line are required. These include replacing pallets, adjusting the parameters for glue application and changing the settings for compression and drying to meet the specific requirements of each product.

The assembly process begins at the first station, where the insulation fabric is carefully placed on a pallet and precisely positioned. A robotic system then transports the pallet together with the fabric to the second station. At this stage, a specialized glue dispensing system applies the glue to the fabric in several places to ensure proper bonding in the subsequent steps.

Next, a metal sheet is carefully positioned on top of the glue-coated fabric. The pallet is then transported to the third station, where pressing and drying operations take place. These steps are crucial to achieve a strong bond between the materials and ensure that the final product meets the required quality standards.

After the bonding process, the product returns to the initial station for final quality inspection. If it meets all specifications, it is placed on a trolley for further processing. How-

ever, products with minor defects are corrected, while products that do not meet quality standards are discarded. Common defects that lead to rejection include contamination on the metal surface, insufficient glue application, visible fingerprints or misalignment of the fabric. As both the glue and the fabric pose potential environmental and health risks, it is essential to switch to more sustainable production methods.

The following components are required for production:

- Sheet metal for Type_N and Type_B
- Insulation fabric for Type_N and Type_B
- Glue

The components are delivered on EU pallets measuring 1200×800 mm. Each pallet contains 1000 sheets. The insulating fabric is supplied in batches of 1000 pieces per pallet for both Type_N and Type_B. Glue is supplied with a maximum of 12,000 ampoules per pallet.

The materials are delivered in kits containing equal quantities of sheet metal, insulating fabric and glue. For example, a shipment of 2000 kits contains 2000 pieces of sheet metal, 2000 pieces of insulating fabric and 2000 ampoules of glue.

The final products are packed in cartons containing 50 products each. These cartons are stacked in sets of 8 per pallet, resulting in a total of 400 finished products per EU pallet (applies to both Type_N and Type_B). All defective products are sent for disassembly; however, this aspect is not covered in this article.

The company follows a strategy in which the components are delivered by three types of vehicle. The one-way distance between the supplier and the company is 32 km:

- Van: with a capacity of 2000 kits or 6 EU pallets of finished products, the van is used for smaller shipments. The average fuel consumption of the van is 11 L of diesel per 100 km and the average transportation cost per trip is €42 (the cost include all expenses incurred during transport, such as fuel, depreciation of the vehicle, drivers' wages, etc.).
- Lorry: The lorry has a capacity of 6000 kits or 18 EU pallets of finished products. Its average fuel consumption is 18 L of diesel per 100 km and the average transportation cost per trip is €68 (the cost include all expenses incurred during transport, such as fuel, depreciation of the vehicle, drivers' wages, etc.).
- Lorry with trailer: The lorry with trailer is used for larger shipments and has a capacity of 10,000 kits or 28 EU pallets of finished products. Its average fuel consumption is 29 L of diesel per 100 km and the average transportation cost per trip is €108 (the cost include all expenses incurred during transport, such as fuel, depreciation of the vehicle, drivers' wages, etc.).

In the following, fuel consumption is highlighted separately from the total transport costs for the purpose of taking into account the sustainability aspect.

Delivery of components is scheduled for 5:30 a.m. on the day on which the stock of components is expected to run out—shortly before the start of production. The finished products are shipped the next morning at 5:40 a.m., the day after production is expected to end. At this time, all finished products stored in the warehouse are shipped.

The company has a total of 16 pallet places—8 for components and 8 for finished products. If additional nearby storage capacity is required, additional pallet storage places can be rented at a price of €0.90 per pallet per week, with a minimum rental period of one week. The rented storage capacity is a tent that is enclosed to protect against weather conditions and is non-temperature controlled warehouse.

4. Digital Twin for Testing System

This section deals with the digital twin of the testing system, which covers the entire supply chain and the entire functioning assembly system and serves as a key tool for performing in-depth analyses. The implementation of a digital twin enables better insight into various processes, optimizes workflows and improves the overall efficiency of the supply chain.

The digital twin was developed based on the logical model of the testing system, shown in Figure 3, using the renowned simulation software Plant Simulation version 2302 [61]. The basic logical model (see Figure 1) has been improved and transformed into a fully functional digital twin (see Figure 3), which is equipped with interactive graphical, numerical and textual outputs to improve usability. The role of each major logical component in a digital twin is explained in Table 1.

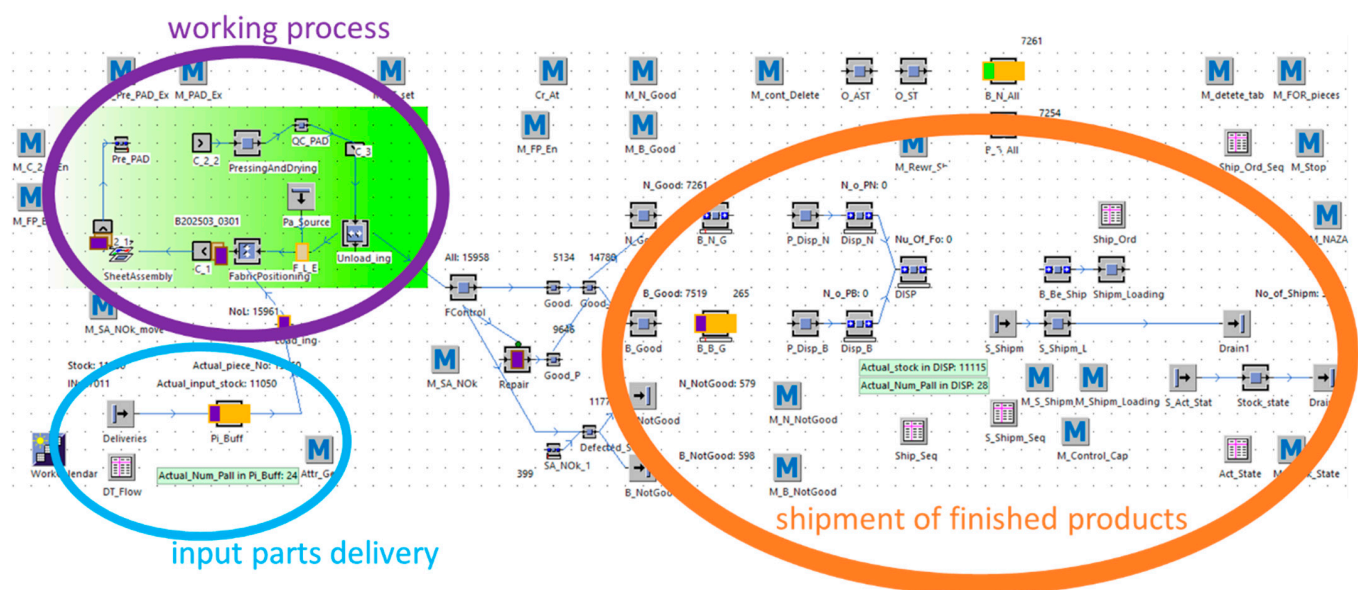


Figure 3. Digital twin for the entire work system, including the delivery of parts and the shipment of finished products with assigned functional segments.

The digital twin consists of three functional segments (Figure 3): The first segment is the “working process” (discussed in detail in Resman et al. [60]), the second segment is the “input parts delivery” and the third segment is the “shipment of finished products”.

The segment of the digital twin “work process” represents a sustainability-oriented digital twin of the testing system, which is discussed in detail in [60]. Its core feature is an embedded algorithm (method M_ST_set in the digital twin) that independently identifies the type of parts within the material flow and assigns the corresponding processing times. The algorithm also automatically recognises changes to the part type within the work system and adjusts the changeover times accordingly to take into account the necessary changes to the work process.

Table 1. Important logical elements in the digital twin for a test work system.

Icon in Figure 3	Description
 DT_Flow	table type objects: DT_Flow: the table with the schedule Ship_Seq: the table with the expected shipping schedule S_Shipm_Seq: the table with the transportation plan Ship_Ord: the table with the data on executed shipments
 EventController	EventController is the object for controlling the simulation process
 SheetAssembly  FControl  PressingAndDrying	process type objects: SheetAssembly (sub-frame): object represent a group of operations Station-type objects represent working processes or objects for decision making Buffer-type objects represent logical element of working process for intermediate storage
 ShiftCalendar	ShiftCalendar type object is used to manage the working calendar and shifts
 	Method type object contains programs for performing logical actions
 Reset	Method type object named Reset delete all movable units in the model and prepare output tables for simulation run
 Init	Method type object Init sets all necessary initial parameters in the model
 Proc_occ	Chart type objects are used to display the efficiency of a manufacturing unit
 Interface	Interface objects are used for logical connections between objects from the main frame and the subframe. The entire SheetAssembly process is modelled in subframe.
	The remaining objects are required to represent all the features of a real production system in the model

4.1. Input Parts Delivery Segment in Digital Twin

The digital twin segment “input parts delivery” comprises the entire delivery policy (Figure 4). The Deliveries object generates the flow of parts into the system based on the delivery sequence data from the DT_Flow table. The Pi_Buff object stores the incoming parts, while the Load_ing object triggers the Attr_Gen method, which contains an algorithm that automatically generates all required part attributes, including their names and types.

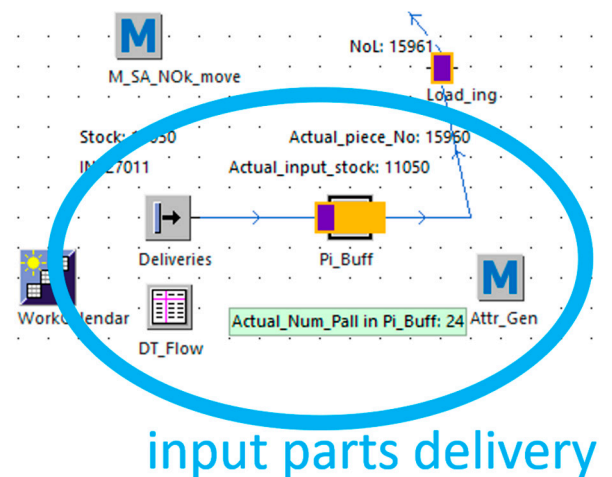


Figure 4. Building objects in the digital twin for the segment “input parts delivery”.

Users only need to enter the delivery data—sequence, part type, quantity and order ID—in the DT_Flow table (an example is shown in Table 2).

Table 2. Example of data in the DT_Flow table in the digital twin.

	Delivery Time	MU	Number	ID_Order
1	5:30:00.0000	.UserObjects.Type_N	866	N202503_0101
2	5:30:01.0000	.UserObjects.Type_B	864	B202503_0102
3	3:05:30:00.0000	.UserObjects.Type_B	967	B202503_0201
4	3:05:30:01.0000	.UserObjects.Type_N	974	N202503_0202
5	10:05:30:00.0000	.UserObjects.Type_N	6000	N202503_0203
...	...	...	...	...

Within the “input parts delivery” segment, a special digital twin algorithm has been developed to manage the flow of parts data. This algorithm is executed via the Attr_Gen method, which is triggered by the Load_ing object. It autonomously generates all the necessary attributes for each part that enters the system and ensures that each attribute is systematically assigned a specific name and data type. This process enables seamless tracking and processing of parts throughout the simulation environment.

4.2. Shipment of Finished Products Segment in Digital Twin

The digital twin segment “shipment of finished products” digital twin segment of the digital twin models covers the entire process of outbound logistics—from the completed production quantities to the execution of customer deliveries (see Figure 5). Once the quality control procedures are completed, the finished parts are sorted into four outbound bins: Acceptable parts are routed to N_Good and B_Good, while rejected parts are stored in N_NotGood and B_NotGood. For each production order, important time-related parameters are monitored autonomously and recorded using special methods (M_N_Good,

M_B_Good, M_N_NotGood and M_B_NotGood), and the resulting data is compiled in the Ship_Seq table.

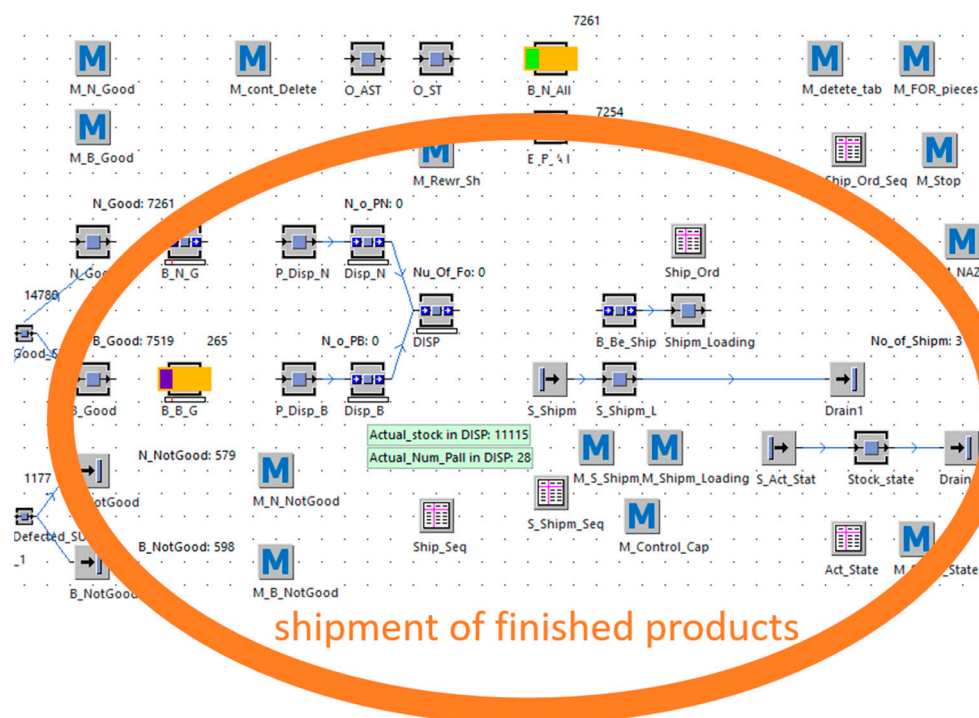


Figure 5. Building objects in the digital twin for the “shipment of finished products” segment.

The quantitative loading of finished parts onto transport pallets is carried out automatically by a special program under the M_Rewr_Sh method. Based on the planned shipping times recorded in Ship_Seq, the arrival of the transporter is initiated via the S_Shipm object. Each individual step in the shipment process—including loading, assignment to a transporter, and shipping—is carried out automatically. Shipment-related data is stored for each order in the Ship_Ord table (see Table 3).

As part of the loading logic, the digital twin also checks the available capacity of the transporter against the quantity of goods to be loaded. If the shipment exceeds the available capacity, the simulation is stopped and the user is notified via a pop-up warning.

There is often a need to monitor real-time or periodic inventory updates—both for incoming raw materials and finished goods. To support this, a custom algorithm has been implemented within the Stock_state object in conjunction with the M_Stock_State method. This function automatically tracks and records the current stock levels and enters them into the Act_State table, either at predefined intervals or when triggered manually.

The logical requirements of the entire working system are fulfilled by developed algorithms that are embedded in the digital twin at appropriate points to enforce the required operating rules.

In this segment of the digital twin, several special algorithms were developed to control and perform key actions:

- An algorithm for autonomous tracking of order completion
- An algorithm for the automatic monitoring of unshipped stock quantities
- An algorithm for the autonomous execution of shipments and the loading of transporters
- A trigger mechanism for an embedded control program that checks transporter capacity and shipment quantities

Table 3. Example of shipping-related data for completed orders in the Ship_ord table in the digital twin.

Seq_n	ID_Order	N_Good	N_All	Start_Pr	Start_Pr_Rel	Finish_Pr	Finish_Pr_Rel	Duration_Pr_Rel	No_Of_Pall	Ship_Time	Ship_Time_Rel
1	N202503_0101	802	866	3 March 2025 06:00:00.0000	5:59:59.0000	4 March 2025 10:35:06.5012	1:10:35:05.5012	1:04:35:06.5012	3	7 March 2025 05:40:01.0000	4:05:40:00.0000
2	B202503_0102	801	864	4 March 2025 10:34:03.8482	1:10:34:02.8482	6 March 2025 10:29:12.6428	3:10:29:11.6428	1:23:55:08.7946	3	7 March 2025 05:40:01.0000	4:05:40:00.0000
3	B202503_0201	895	967	6 March 2025 10:28:03.2907	3:10:28:02.2907	11 March 2025 06:54:44.2916	8:06:54:43.2916	4:20:26:41.0009	3	14 March 2025 05:40:01.0000	11:05:40:00.0000
4	N202503_0202	902	974	11 March 2025 06:53:42.7257	8:06:53:41.7257	13 March 2025 07:46:44.8440	10:07:46:43.8440	2:00:53:02.1183	3	14 March 2025 05:40:01.0000	11:05:40:00.0000
5	N202503_0203	5557	6000	13 March 2025 07:45:11.6516	10:07:45:10.6516	2 April 2025 06:22:36.6155	30:06:22:35.6155	19:22:37:24.9639	14	22 April 2025 05:40:01.0000	50:05:40:00.0000
6	B202503_0204	5558	6000	2 April 2025 06:21:15.1634	30:06:21:14.1634	21 April 2025 09:19:54.0805	49:09:19:53.0805	19:02:58:38.9171	14	22 April 2025 05:40:01.0000	50:05:40:00.0000
7	B202503_0301	10,504	11,340	21 April 2025 09:18:23.3504	49:09:18:22.3504	27 May 2025 08:05:14.9252	85:08:05:13.9252	35:22:46:51.5748	27	28 May 2025 05:40:01.0000	86:05:40:00.0000
8	N202503_0302	10,503	11,340	27 May 2025 08:03:54.8970	85:08:03:53.8970	2 July 2025 07:17:56.6591	121:07:17:55.6591	35:23:14:01.7621	27	3 July 2025 05:40:01.0000	122:05:40:00.0000
9	N202503_0401	11,115	12,000	2 July 2025 07:16:38.4819	121:07:16:37.4819	8 August 2025 08:43:12.8392	158:08:43:11.8392	37:01:26:34.3573	28	9 August 2025 05:40:01.0000	159:05:40:00.0000
10	N202503_0402	2890	3120	8 August 2025 08:41:36.7969	158:08:41:35.7969	19 August 2025 09:04:05.1601	169:09:04:04.1601	11:00:22:28.3632	8	20 August 2025 05:40:01.0000	170:05:40:00.0000
11	B202503_0403	2890	3120	19 August 2025 09:02:40.9537	169:09:02:39.9537	28 August 2025 09:22:50.1726	178:09:22:49.1726	9:00:20:09.2189	8	29 August 2025 05:40:01.0000	179:05:40:00.0000
...	...	...	...	...	...	...	...	...	...	...	...

Legend: ID_Order—ID number of the order; N_good—number of good parts in the finished order; N_all—number of input parts in the order; Start_Pr—start of the production time of the order in date format; Start_Pr_Rel—start of the production time of the order in absolute time format; Finish_Pr—end of the production time of the order in date format; Finish_Pr_Rel—end of the production time of the order in absolute time format; Duration_Pr_rel—duration of the production time of the order in absolute time format; No_Of_Pall—number of pallets (with loaded pieces) in the finished order; Ship_time—start of the shipment of the order in date format; Ship_time_rel—start of the shipment time of the order in absolute time format.

4.2.1. Algorithm for Autonomous Tracking of Order Completion

In order to enable autonomous and continuous tracking of completion times for each individual order, a separate algorithm was developed.

For each order, the time values are automatically monitored and recorded using the M_N_Good, M_B_Good, M_N_NotGood and M_B_NotGood methods (as shown in Figure 5), which are triggered after quality control is completed for each individual part.

Once the last part of an order has been completed, the time data for this order is collected and saved under the corresponding order attributes.

The final execution step of the algorithm is triggered by the M_Rewr_Sh (Algorithm 1) method, which writes the output data for each order to the Ship_Seq table.

Algorithm 1. Part of the algorithm for autonomous tracking of order completion

```

var sto_501, sto_503, sto_504, sto_511 : integer;
var sto_502 : string;
var sto_505, sto_507, sto_509 : datetime;
var sto_506, sto_508, sto_510 : time;
...

vrsta := str_to_num(.Models.Model.DISP.Label);

sto_502 := @.getAttrValue(2);
sto_503 := @.getAttrValue(3);
sto_504 := .Models.Model.DT_Flow [3,vrsta];
sto_505 := @.getAttrValue(5);
sto_506 := @.getAttrValue(6);
sto_507 := @.getAttrValue(7);
sto_508 := @.getAttrValue(8);
sto_509 := @.getAttrValue(9);
sto_510 := @.getAttrValue(10);

sto_501 := vrsta;

.Models.Model.Ship_Seq[1,vrsta] := num_to_str(sto_501);
.Models.Model.Ship_Seq[2,vrsta] := sto_502;
.Models.Model.Ship_Seq[3,vrsta] := num_to_str(sto_503);
.Models.Model.Ship_Seq[4,vrsta] := num_to_str(sto_504);
.Models.Model.Ship_Seq[5,vrsta] := datetime_to_str(sto_505);
.Models.Model.Ship_Seq[6,vrsta] := time_to_str(sto_506);
.Models.Model.Ship_Seq[7,vrsta] := datetime_to_str(sto_507);
.Models.Model.Ship_Seq[8,vrsta] := time_to_str(sto_508);
.Models.Model.Ship_Seq[9,vrsta] := datetime_to_str(sto_509);
.Models.Model.Ship_Seq[10,vrsta] := time_to_str(sto_510);

sto_511 := (sto_503 div 400) + 1;
.Models.Model.Ship_Seq[11,vrsta] := num_to_str(sto_511);

@.setAttrValue(4,sto_504);
@.setAttrValue(11,sto_511);

```

Algorithm 1. *Cont.*

```

.Models.Model.DISP.Label := num_to_str(vrsta + 1);

// print of actual state of pieces and palletes in output storage
if B_N_G.occupied then
  ASK_N := B_N_G.numMU;
  ASP_N := (B_N_G.numMU div 400) + 1;
else
  ASK_N := 0;
  ASP_N := 0;
end;

if B_B_G.occupied then
  ASK_B := B_B_G.numMU;
  ASP_B := (B_B_G.numMU div 400) + 1;
else
  ASK_B := 0;
  ASP_B := 0;
end;

stanje_prej := str_to_num(Deliveries.Label);
stanje_aktual := stanje_prej + sto_503 + ASK_N + ASK_B; -- actual state of pieces
pal_prej := str_to_num(S_Act_Stat.Label);
pal_aktual := pal_prej + sto_511 + ASP_N + ASP_B; -- actual state of pallets
Comment2.Text := "Actual_stock in DISP: " + num_to_str(stanje_aktual);
Comment3.Text := "Actual_Num_Pall in DISP: " + num_to_str(pal_aktual);
Deliveries.Label := num_to_str(stanje_aktual);
S_Act_Stat.Label := num_to_str(pal_aktual);

// end of actual state of pieces and palletes in output storage

```

4.2.2. Algorithm for Automatic Stock Tracking of Unshipped Quantities

To enable autonomous and continuous tracking of unshipped stock quantities, a special algorithm was developed to monitor the number of finished parts and the corresponding pallet stock.

The algorithm is implemented in the M_Stock_State method (as shown in Figure 5).

This method works with the Stock_state object and automatically enters the current stock status in the Act_State table at predefined intervals.

In addition, the M_Stock_State method can be triggered manually to perform data recording as required.

4.2.3. Algorithm for the Autonomous Execution of Shipments and the Loading of Transporters

A special algorithm was developed to automate the loading of the transporters and record the shipment data in a specific table in order to enable the autonomous execution of shipments.

The shipment sequence is predefined in the Ship_Seq table (as shown in Figure 5), in which both the planned shipment time and the transporter type are specified for each entry.

The shipments are triggered autonomously through the S_Shipm object, which also ensures that a transporter is available for loading.

Loading takes place at the S_Shipm_L object, where the logical loading of the pending pallets from the warehouse (DISP object) is carried out. This step includes the automatic execution of embedded algorithms within the methods M_Shipm and M_Shipm_Loading.

These methods not only control the loading process, but also perform tracking functions for each shipment. As a result, shipment data is collected and recorded in the Ship_Ord table.

4.2.4. Control Function for Verifying Transporter Capacities and the Quantities to Be Shipped

A special control function was developed and integrated into the algorithm described above to autonomously check the transporter capacities and the quantities intended for dispatch.

This control function is implemented in the M_Control_Cap method (as shown in Figure 5). As a result, a capacity check is carried out for each loading process. In the event of overcapacity, the simulation is automatically stopped and the user is informed of the problem via a pop-up notification.

4.3. Validation and Verification

Once the development of the digital twin was complete, a thorough validation and verification process was carried out. This critical phase utilized a comprehensive data set obtained from historical records of deliveries, manufacturing activities and shipments of completed orders of the actual operating system.

The test data spanned several periods over the past year and included the following order sets: a batch of five orders totaling 2000 finished goods, a batch of seven orders totaling 3100 finished goods, and a final batch of four orders totaling 880 finished goods.

This diverse data set was deliberately chosen to represent the typical operational variations and quantity fluctuations of the system and to ensure a robust assessment of the accuracy and reliability of the digital twin in different scenarios.

5. Results

This section presents the results of the analysis conducted using the developed digital twin. As part of the study, four different tests were performed to evaluate the impact of the different transport configurations on system performance.

The tests are defined as follows:

- TEST_1: Focuses on the transport of materials and the shipment of finished products using vans only.
- TEST_2: Focuses on the transport of materials and the shipment of finished products using lorries only.
- TEST_3: Focuses on the transportation of materials and shipment of finished products using only lorries with trailer.
- TEST_4: Focuses on transporting materials and shipping finished products using a mix of all three types of transport vehicles (vans, lorries and lorries with trailer).

In all tests, the order quantities remained constant to ensure that the differences observed in the results were exclusively due to the type of vehicle used in logistics. The order quantities are as follows:

- Order 1 requires the production of 800 units of Type_N and 800 units of Type_B.
- Order 2 requires the production of 6300 units of Type_N and 6300 units of Type_B.
- Order 3 requires 10,500 units of Type_N and 10,500 units of Type_B.

- Order 4 requires 14,000 units of Type_N and 14,000 units of Type_B.

By maintaining consistent production requirements across all tests, we can accurately assess how different transportation methods affect efficiency in both component deliveries and the shipping of finished products.

The recorded quantity of each order reflects the total number of products that the company must produce to satisfy customer demand. However, during the production process, a certain percentage of defective products will inevitably be produced. These defective parts are discarded and directed toward recycling. In order to account for possible defects and ensure that production targets are met without interruption, the supply quantities of raw materials have been increased for the amount of expected defective finished products (approximately 9%).

For all tests, data collection began on 3 March 2025 at 06:00:00. It was assumed that the production system is in operation from Monday to Friday from 06:00:00 to 10:40:00, with two 20-min breaks in between.

5.1. Analysis of Incoming and Outgoing Logistics and Warehouse Utilization

With the help of the digital twin, we were able to create detailed recommendations for the scheduling of component deliveries and the shipment of finished products. These recommendations help to ensure a smooth and efficient production flow by synchronizing the arrival of deliveries with manufacturing needs while optimizing the distribution of finished products.

In our specific case, the proposed schedule specifies the timing and frequency of both component deliveries and shipment of the final product, helping to minimize delays, reduce inventory costs and improve overall logistical efficiency. The detailed schedule for the delivery of components and shipment of products is shown in Table 4 and provides a clear overview of the planned processes.

All deliveries and shipments of finished products within the company are planned before the start of the production day. This strategic approach is designed to efficiently manage logistical challenges and ensure a smooth workflow without disrupting the manufacturing process. If an order is completed in the middle of a working day, the components for the next production cycle are delivered on the same day—before the start of production. This ensures an uninterrupted workflow and the maintenance of operational continuity.

Tables 4–7 provide a detailed overview of the delivery schedule, broken down by individual customer orders and the type of transport vehicle used. In the first test case, the entire inbound and outbound delivery logistics are handled by vans. In order to meet the requirements of all four orders, a total of 36 van trips are required for incoming materials and a further 36 trips for the delivery of finished products to customers. This results in a total of 72 van trips.

Order 1 follows a relatively straightforward delivery schedule. All materials required for production are delivered in a single transport on the first day, with the van arriving on site at 5:30 a.m. Production begins shortly afterwards. The finished products for order 1 are scheduled for shipment on the morning of day 4, ensuring quick delivery to the customer shortly after production begins.

Order 2 follows an extended and segmented delivery plan. The material is delivered in seven separate transports, starting at 5:30 a.m. on day 3. These staggered delivery series support a continuous flow of components into production. Production for order 2 extends over a longer period and ends on day 50, when the last batch of finished products is scheduled for shipment.

Table 4. Supply chain of components (quantity of components) for product manufacturing (required quantity of good parts), including delivery and shipping times in the case of transportation by van (TEST_1).

Order ID	Quantity of Good Products	Type_N Components	Type_B Components	Delivery Time [DD:HH:SS]	Type of Vehicle for Delivery	Shipping Time [DD:HH:SS]	Type of Vehicle for Shipping
202503_01	800 Type N 800 Type B	866	864	00:05:30:00	Van	4:05:40:00.00	Van
202503_02	6300 Type N 6300 Type B	2000	0	03:05:30:00	Van	11:05:40:00.0000	Van
		2000	0	10:05:30:00	Van	18:05:40:00.0000	Van
		2000	0	17:05:30:00	Van	24:05:40:00.0000	Van
		967	974	23:05:30:00	Van	31:05:40:00.0000	Van
		0	2000	30:05:30:00	Van	37:05:40:00.0000	Van
		0	2000	36:05:30:00	Van	44:05:40:00.0000	Van
		0	2000	43:05:30:00	Van	50:05:40:00.0000	Van
202503_03	10,500 Type N 10,500 Type B	0	2000	49:05:30:00	Van	57:05:40:00.0000	Van
		0	2000	56:05:30:00	Van	61:05:40:00.0000	Van
		0	2000	60:05:30:00	Van	68:05: 40:00.0000	Van
		0	2000	67:05:30:00	Van	74:05: 40:00.0000	Van
		0	2000	73:05:30:00	Van	81:05: 40:00.0000	Van
		0	1340	80:05:30:00	Van	86:05: 40:00.0000	Van
		2000	0	85:05:30:00	Van	92:05: 40:00.0000	Van
		2000	0	91:05:30:00	Van	99:05: 40:00.0000	Van
		2000	0	98:05:30:00	Van	106:05:40:00.0000	Van
		2000	0	105:05:30:00	Van	110:05:40:00.0000	Van
		2000	0	109:05:30:00	Van	117:05:40:00.0000	Van
		1340	0	116:05:30:00	Van	122:05:40:00.0000	Van
202503_04	14,000 Type N 14,000 Type B	2000	0	121:05:30:00	Van	128:05:40:00.0000	Van
		2000	0	127:05:30:00	Van	135:05:40:00.0000	Van
		2000	0	134:05:30:00	Van	141:05:40:00.0000	Van
		2000	0	140:05:30:00	Van	148:05:40:00.0000	Van
		2000	0	147:05:30:00	Van	152:05:40:00.0000	Van
		2000	0	151:05:30:00	Van	159:05:40:00.0000	Van
		2000	0	158:05:30:00	Van	166:05:40:00.0000	Van
		1120	0	165:05:30:00	Van	170:05:40:00.0000	Van
		0	2000	169:05:30:00	Van	177:05:40:00.0000	Van
		0	2000	176:05:30:00	Van	183:05:40:00.0000	Van
		0	2000	182:05:30:00	Van	190:05:40:00.0000	Van
		0	2000	189:05:30:00	Van	194:05:40:00.0000	Van
		0	2000	193:05:30:00	Van	201:05:40:00.0000	Van
		0	2000	200:05:30:00	Van	207:05:40:00.0000	Van
		0	2000	206:05:30:00	Van	214:05:40:00.0000	Van
		0	1120	213:05:30:00	Van	218:05:40:00.0000	Van

Orders 3 and 4 have the largest volume of the four orders and require more complex transport coordination. Order 3 includes a total of 24 van movements—12 for incoming material deliveries and 12 for outgoing shipments. Order 4, the most logistically demanding, requires 32 van movements, evenly distributed between deliveries and shipments. These two orders have a higher degree of production complexity and logistical planning due to the larger quantities involved.

The production time for all four orders is 218 days. This extended timeframe reflects the cumulative production and delivery requirements for all orders. The final shipment of finished goods, marking the completion of all production activities, is scheduled for 5:30 a.m. on day 218. The carefully coordinated schedule ensures that all customer orders are fulfilled on time, with efficient use of transportation and minimal disruption to production operations.

Table 5. Supply chain of components (quantity of components) for product manufacturing (required quantity of good parts), including delivery and shipping times for transport by lorry (TEST_2).

Order ID	Quantity of Good Products	Type_N Components	Type_B Components	Delivery Time [DD:HH:SS]	Type of Vehicle for Delivery	Shipping Time [DD:HH:SS]	Type of Vehicle for Shipping
202503_01	800 Type N 800 Type B	866	864	00:05:30:00	Lor	04:05:40:00	Lor
202503_02	6300 Type N 6300 Type B	6000	0	03:05:30:00	Lor	24:05:40:00	Lor
		967	974	23:05:30:00	Lor	31:05:40:00	Lor
		0	6000	30:05:30:00	Lor	50:05:40:00	Lor
202503_03	10,500 Type N 10,500 Type B	0	6000	49:05:30:00	Lor	68:05:40:00	Lor
		0	5340	67:05:30:00	Lor	86:05:40:00	Lor
		6000	0	85:05:30:00	Lor	106:05:40:00	Lor
		5340	0	105:05:30:00	Lor	122:05:40:00	Lor
202503_04	14,000 Type N 14,000 Type B	6000	0	121:05:30:00	Lor	141:05:40:00	Lor
		6000	0	140:05:30:00	Lor	159:05:40:00	Lor
		3120	0	158:05:30:00	Lor	170:05:40:00	Lor
		0	3120	169:05:30:00	Lor	179:05:40:00	Lor
		0	6000	178:05:30:00	Lor	199:05:40:00	Lor
		0	6000	198:05:30:00	Lor	218:05:40:00	Lor

Table 6. Supply chain of components (quantity of components) for product manufacturing (required quantity of good parts), including delivery and shipping times for transport by lorry with trailer (TEST_3).

Order ID	Quantity of Good Products	Type_N Components	Type_B Components	Delivery Time [DD:HH:SS]	Type of Vehicle for Delivery	Shipping Time [DD:HH:SS]	Type of Vehicle for Shipping
202503_01	800 Type N 800 Type B	866	864	00:05:30:00	Lor_T	04:05:40:00	Lor_T
202503_02	6300 Type N 6300 Type B	6967	0	03:05:30:00	Lor_T	26:05:40:00	Lor_T
		0	6974	25:05:30:00	Lor_T	50:05:40:00	Lor_T
202503_03	10,500 Type N 10,500 Type B	0	11,340	49:05:30:00	Lor_T	86:05:40:00	Lor_T
		11,340	0	85:05:30:00	Lor_T	122:05:40:00	Lor_T
202503_04	14,000 Type N 14,000 Type B	12,000	0	121:05:30:00	Lor_T	159:05:40:00	Lor_T
		3120	3120	158:05:30:00	Lor_T	179:05:40:00	Lor_T
		0	12,000	158:05:30:01	Lor_T	218:05:40:00	Lor_T

Table 5 (TEST_2) illustrates the distribution of component deliveries and the shipment of finished products exclusively by lorry. Compared to vans, significantly fewer trips are required when using lorries due to their higher capacity. In total, only 14 trips are needed to deliver all the components required for production, and a further 14 trips are sufficient to ship the finished products for all four orders.

Order 4 requires the highest number of deliveries, which is to be expected given the large production volume. The reduced number of trips underscores the efficiency of using lorries in large-scale logistics, as they enable the consolidation of materials and products into fewer shipments with greater capacity. This optimized delivery schedule not only reduces the number of transport operations, but also simplifies logistics coordination and has the potential to reduce overall transport costs.

Table 7. Supply chain of components (quantity of components) for product manufacturing (required quantity of good parts), including delivery and shipping times in the case of transport using all three types of vehicles (TEST_4).

Order ID	Quantity of Good Products	Type_N Components	Type_B Components	Delivery Time [DD:HH:SS]	Type of Vehicle for Delivery	Shipping Time [DD:HH:SS]	Type of Vehicle for Shipping
202503_01	800 Type N 800 Type B	866	864	00:05:30:00	Van	04:05:40:00	Van
202503_02	6300 Type N 6300 Type B	967 6000	974 6000	03:05:30:00 10:05:30:00	Van Lor_T	11:05:40:00 50:05:40:00	Lor_T Lor_T
202503_03	10,500 Type N 10,500 Type B	0 11,340	11,340 0	49:05:30:00 85:05:30:00	Lor_T Lor_T	86:05:40:00 122:05:40:00	Lor_T Lor_T
202503_04	14,000 Type N 14,000 Type B	12,000	0	121:05:30:00	Lor_T	159:05:40:00	Lor_T
		3120	0	158:05:30:00	Lor	170:05:40:00	Lor_T
		0	3120	169:05:30:00	Lor	179:05:40:00	Lor
		0	12,000	178:05:30:00	Lor_T	218:05:40:00	Lor_T

An even lower number of trips is required when using a lorry with trailer, as shown in Table 6 (TEST_3). Due to the significantly larger loading capacity, the total number of trips is reduced to only 8 for the delivery of production components and another 8 for the shipment of finished products, covering all four customer orders.

This further optimization in transport logistics underlines the efficiency gains that can be achieved with higher-capacity vehicles. By consolidating larger volumes into fewer trips, the use of lorries with trailer not only reduces the total number of transport operations, but also minimizes handling time, lowers traffic impact, and simplifies overall logistics coordination.

If all vehicle types are combined, the sequence of deliveries and shipments changes, as shown in Table 7 (TEST_4).

By using the capabilities of a digital twin, it is possible to conduct a detailed analysis of storage space requirements throughout the production process, which is particularly important given the company's limited pallet storage capacity. One focus of this analysis is to determine the number of pallet places needed to ensure smooth and uninterrupted operations—both when receiving raw materials and when shipping finished products.

As mentioned above, the company has a fixed internal capacity of 8 pallet places dedicated to the delivery and temporary storage of incoming components and a further 8 pallet storage places reserved for finished goods awaiting shipment. While this internal capacity is generally sufficient, additional pallet storage places must be rented from an external storage provider if there is a need for more.

In order to manage this efficiently, the company has concluded a contract with a logistics partner that enables the weekly rental of pallet places. This agreement offers the flexibility to expand storage capacity in line with weekly demand. However, accurate forecasting is essential to avoid both overbooking and under-utilisation.

By simulating the entire production schedule, delivery times and shipping requirements, the digital twin calculates the exact number of pallet places needed each week. It takes into account the timing and volume of component deliveries, production rates and shipping dates, providing a dynamic and data-driven approach to warehouse space planning.

With this information, decision-makers can proactively reserve the required pallet storage place in advance and thus ensure uninterrupted production, even at peak times. At the same time, the company avoids unnecessary costs by not renting more places than necessary. The integration of the digital twin into warehouse planning enables smarter

logistics management, improved cost efficiency and greater responsiveness to fluctuations in production demand.

The external warehouse service provider charges a weekly rental fee of € 0.90 per pallet place. As part of the analysis, fuel consumption and transportation costs were also evaluated across all tests.

Figure 6 shows the weekly demand for pallet places for incoming components and outgoing finished products during the entire production period for the first test case (TEST_1). This data is important to understand the dynamics of storage needs over time. In the diagram, a red line indicates the company's available internal storage capacity. This reference line can be used to assess whether the forecasted demand for pallet storage locations exceeds the on-site capacity in a given week.

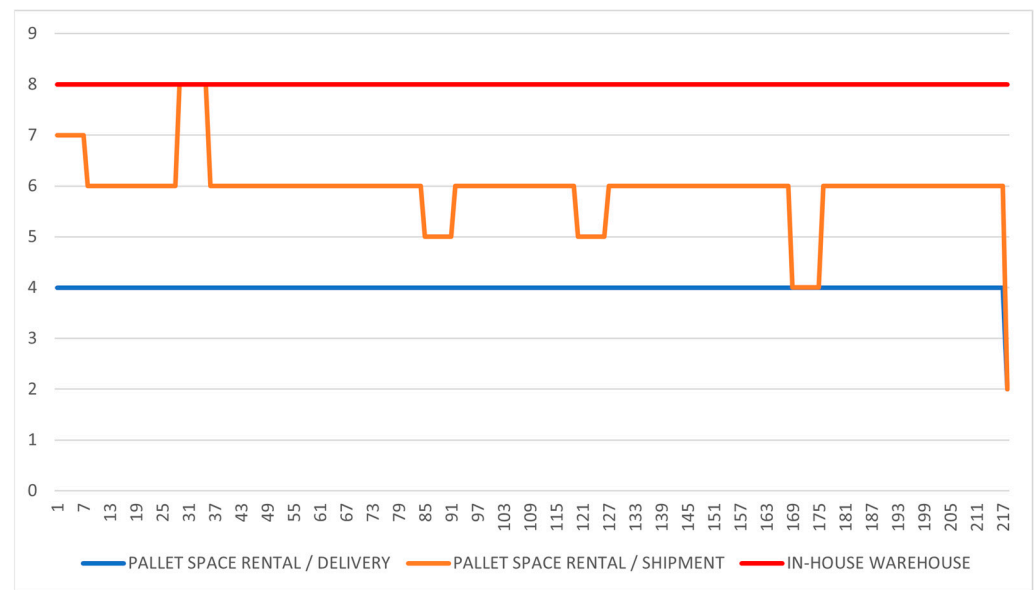


Figure 6. Weekly dynamics of the required pallet storage places for components and finished products in TEST_1.

The graph clearly shows that the weekly demand for pallet spaces is always within the limits of internal storage capacity when vans are used to transport both components and finished products.

In TEST_2, the situation is completely different, as shown in Figure 7. In contrast to TEST_1, this transport method leads to a pallet storage place requirement that exceeds the company's internal storage capacity and makes it necessary to rent additional pallet places for both the incoming components and the finished products.

According to the data shown in the figure, the need for external storage becomes apparent at an early stage of the production process. The peak demand for incoming components reaches up to 12 pallet places in a single week—four more than the company's internal limit. This shows that in certain periods, up to 50% of the required pallet spaces for components have to be sourced from external warehouse providers.

The situation is even more pronounced for finished products. The weekly peak demand for the storage of outgoing goods reaches 15 pallet storage places, which is almost double the internal capacity. This clearly shows that while lorries offer greater transport efficiency and require fewer trips, they can present logistical challenges related to storage, particularly when delivery and shipping schedules are not closely aligned with production.

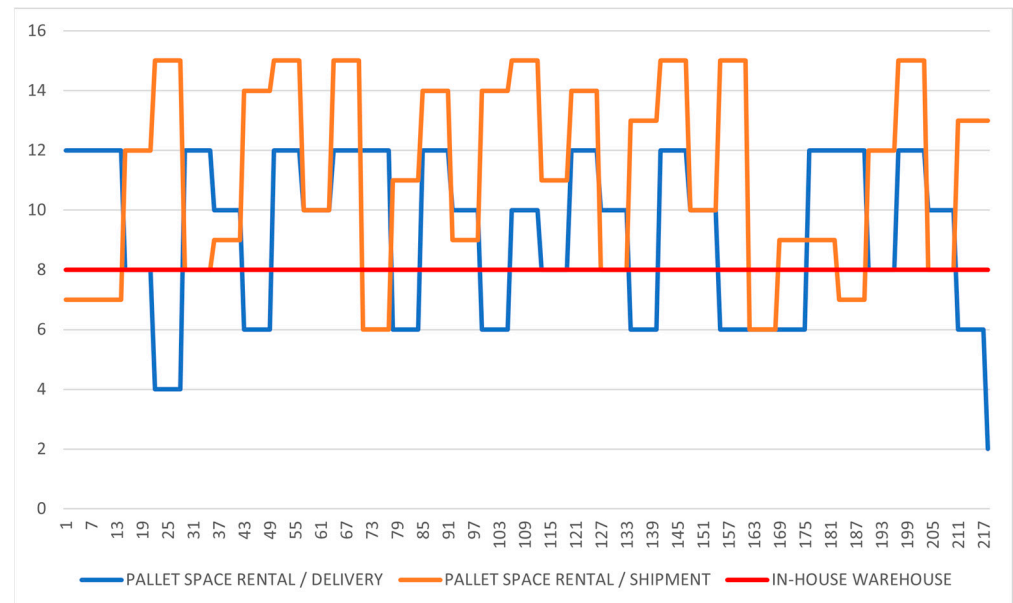


Figure 7. Weekly dynamics of the required pallet storage places for components and finished products in TEST_2.

An even larger number of pallet storage places must be rented in TEST_3 if a lorry with trailer is used for transport, as shown in Figure 8 and Table 6. Although this transport variant offers the highest transport capacity and requires the lowest number of trips, it leads to considerable peaks in demand for pallet storage places due to the large quantities that are delivered and dispatched at once. As a result, the internal storage capacity is quickly exceeded, making it necessary to rent additional warehouse space on a much larger scale.

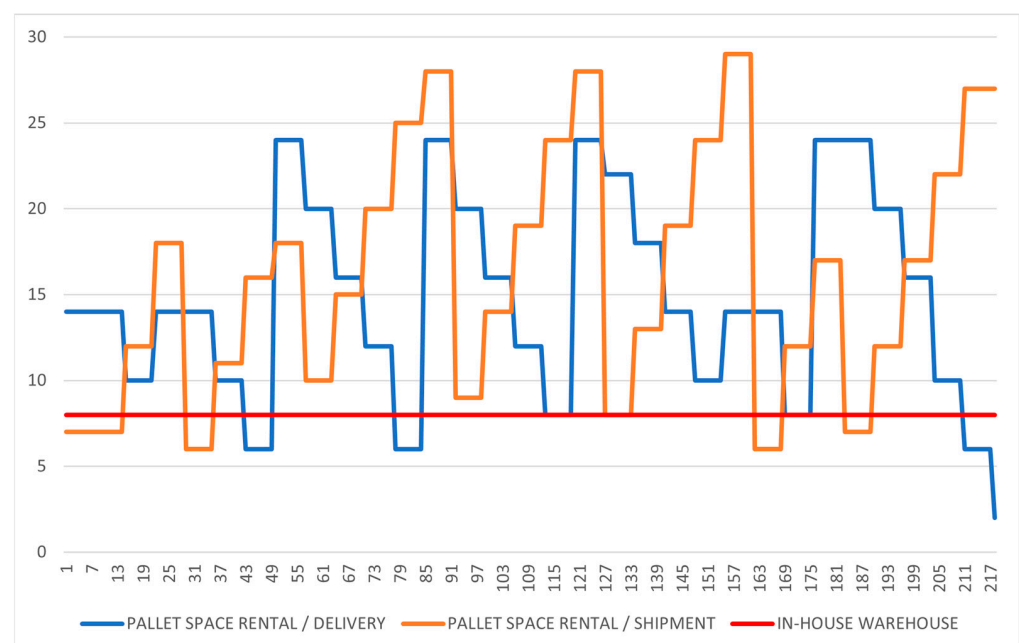


Figure 8. Weekly dynamics of the required pallet storage places for components and finished products during transport in TEST_3.

For finished products, the weekly demand for pallet spaces can reach up to 28 places—more than three times the internal capacity provided for outgoing products. This sharp

increase results from the lower frequency but higher volume of shipments, leading to a temporary accumulation of finished goods awaiting shipment.

The demand for pallet storage places for incoming components is also increasing significantly. At its peak, the demand reaches 24 pallet positions, which in turn far exceeds the internal capacity of 8 pallets for incoming materials. This is a direct consequence of the ability of the lorry with trailer to transport large quantities in a single trip.

A comparable volume of rented pallet storage places is also required in the TEST_4 case (Table 7 and Figure 9). This approach, which aims to balance the advantages of each mode of transport—vans, lorries and lorries with trailer—still leads to significant peaks in storage demand due to the large and irregular deliveries associated with higher capacity vehicles.

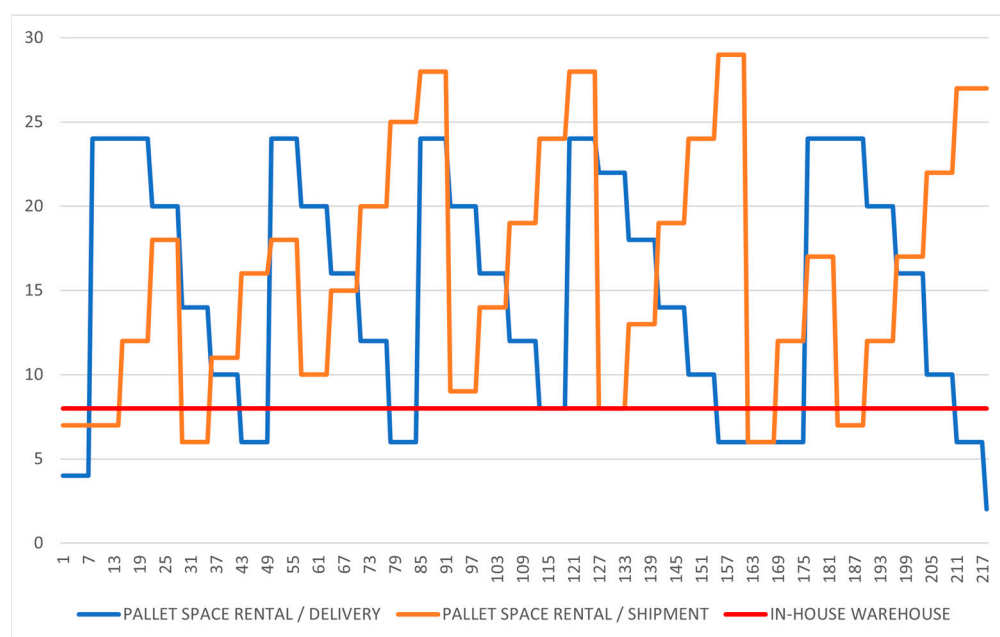


Figure 9. Weekly dynamics of the required pallet storage places for components and finished products in TEST_4.

5.2. Cost Analysis of Transportation and Warehousing

This subsection presents the cost results for all four test cases, obtained using the digital twin.

In the case of TEST_1, which involves the transportation of components and finished products using vans (Figure 10), the total transport costs amounted to €2808, with a corresponding fuel consumption of around 253 L. A major advantage of the van is the elimination of external warehousing, so that no costs are incurred for renting pallet places.

In the case of TEST_2 (Figure 11), the values obtained from the simulation of the digital twin differ from those observed for transportation with van. The fuel consumption for this type of transport is around 161 L, while the total transportation costs amount to €1932.

As the volume of products delivered and shipped exceeds the company's internal pallet storage capacity, external pallet storage places have to be rented. The associated rental costs amount to around €390 for incoming components and €693 for finished products.

In the case of TEST_3 (Figure 12), the results of the digital twin simulation show further differences. The fuel consumption for this transport mode is about 148 L and the total transportation cost is €1680. As the quantity of components delivered and finished products shipped significantly exceeds the internal pallet storage capacity, it becomes necessary to

rent an external warehouse. The costs for renting pallet storage places amount to around €1398 for incoming components and around €1751 for finished products.

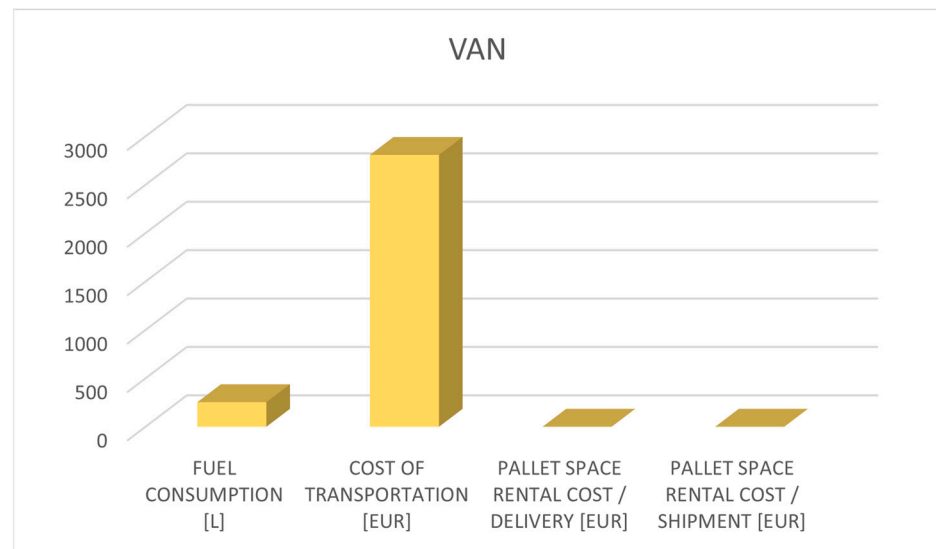


Figure 10. Fuel consumption and associated costs for the transport of components and finished products in TEST_1.

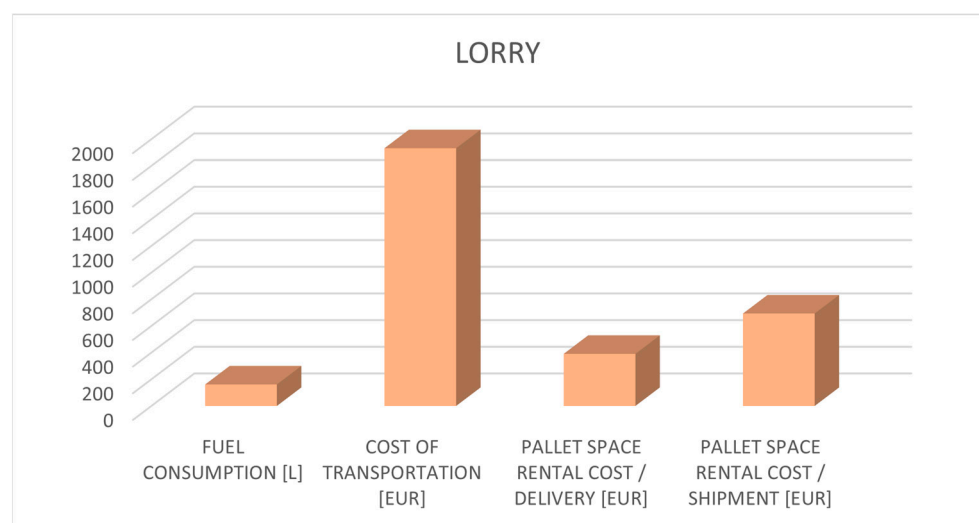


Figure 11. Fuel consumption and associated costs for the transport of components and finished products in TEST_2.

The simulation of a combined transportation strategy—TEST_4 (Figure 13) shows the following key results: The fuel consumption for this multimodal transport is around 139 L, and the total transportation costs are calculated to be around €1626.

A critical factor identified in the simulation is the considerable volume of both incoming components and outgoing finished goods. This volume exceeds the company's internal pallet storage capacity throughout the entire transport chain, making the use of external warehouses unavoidable. The cost of renting pallet storage places therefore amounts to around €1474 for incoming components and around €1902 for finished products.

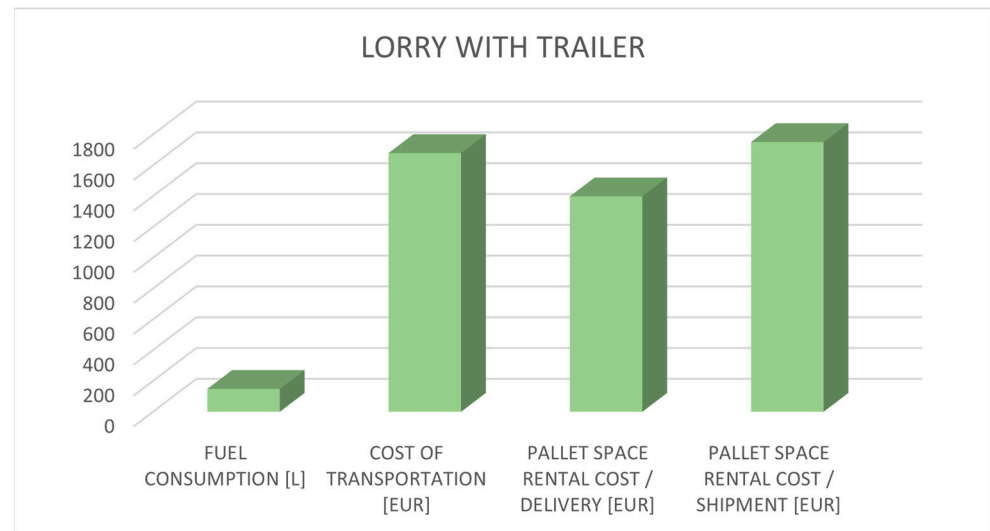


Figure 12. Fuel consumption and associated costs for the transport of components and finished products in TEST_3.

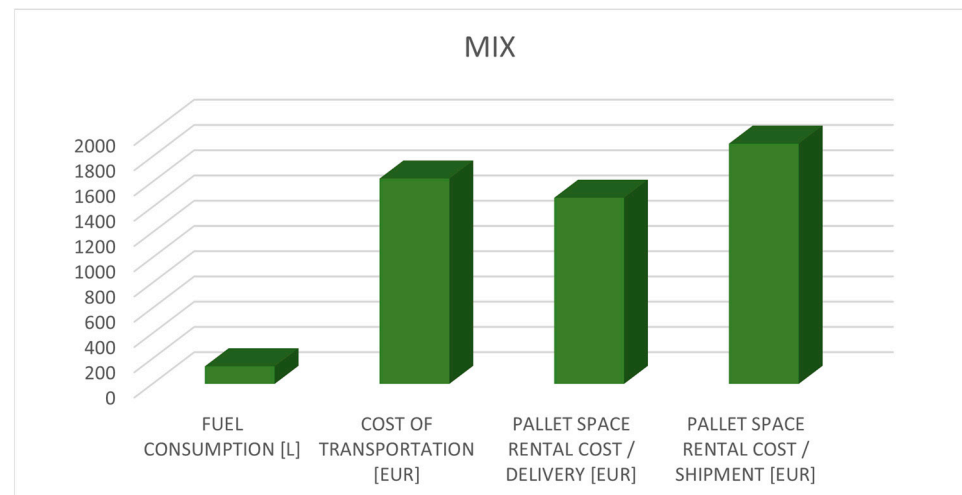


Figure 13. Fuel consumption and associated costs for the transport of components and finished products in TEST_4.

6. Discussion

The developed approach, based on a highly adaptable digital twin, provides a robust platform for comprehensive analysis and strategic optimization in both logistics and production environments. Its ability to process different order sets at the individual part level enables granular demand fulfillment simulations and efficient resource allocation. In addition, the digital twin's ability to manage different shipping configurations supports accurate outbound logistics modeling and advanced transportation planning. This detailed approach accommodates a broad range of unit volumes within individual orders and provides a high degree of flexibility when analyzing operational scenarios with fluctuating demand.

As part of the development of the methodology and the implementation of the supporting algorithms, over 850 lines of program code were written within the digital twin to address all specific aspects of the defined problem. This ensured the functionality and adaptability of the digital twin required to perform advanced simulation and optimization

tasks. This makes the solution ideal for testing and analyzing supply chains in a variety of discrete manufacturing modes.

The applicability of the research results presented was demonstrated through testing and the necessary simulation runs, which included scenarios with different types of transportation units—vans, lorries, lorries with trailers and a combined approach—and evaluated against key logistical performance indicators such as fuel consumption and storage requirements. A key finding was the inherent trade-off between transportation frequency and storage requirements. The use of vans resulted in the highest number of transport operations (72 in total, as shown in Table 4) due to their limited loading capacity (Figure 6). However, this allowed the company to fully utilize the available internal warehouse space and avoid external storage of pallets. Despite the highest fuel consumption and the highest total transportation costs (€2808), this scenario did not incur any pallet rental costs. Therefore, it proved to be the most storage-efficient option that was closely aligned with the company’s existing infrastructure and supported a streamlined logistics flow (Figure 14).

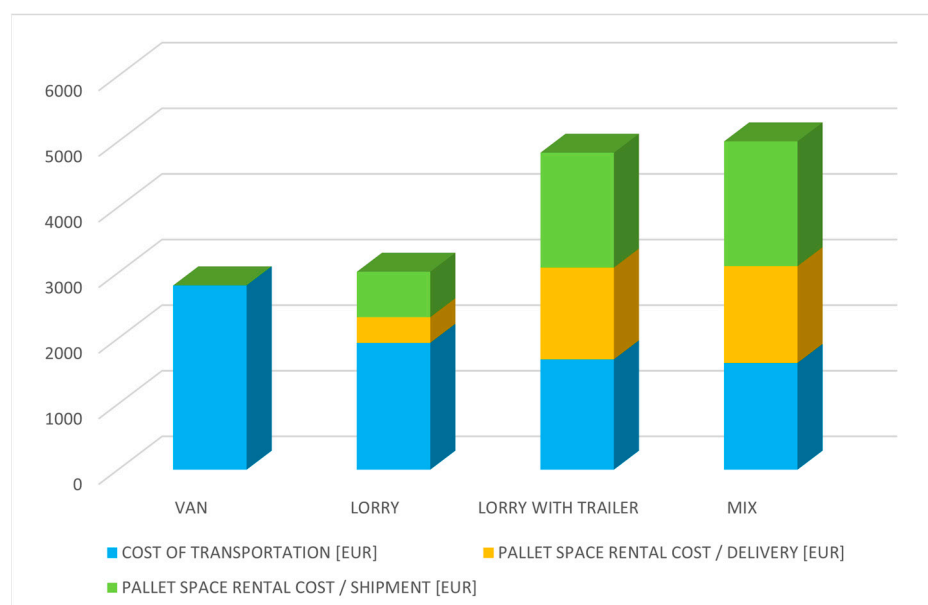


Figure 14. Cost comparison between different types of transport strategy, defined in TEST_1 to TEST_4.

The use of lorries, on the other hand, significantly reduced the number of trips required to just 28, which represents a considerable increase in transport efficiency. Fuel consumption also decreased and transportation costs fell to €1932 (Figure 14). However, warehouse demand exceeded internal capacity and additional pallet places had to be rented. The total costs for the TEST_2 case therefore amounted to around €3015 (Figure 14).

Although the TEST_3 case involved only 16 trips, it led to considerable peaks in weekly demand for pallet storage places—up to 24 for incoming materials and 28 for finished goods (Table 6). Although the transportation costs were slightly lower at €1680, the rental of pallet storage spaces contributed significantly to the total logistics costs (around €4830) (Figure 14).

The tests have shown that the total costs for TEST_4 case are the highest (about €5003), although the transportation costs are the lowest in this case. Most of the costs are related to the rental of external warehouses, as shown in Figure 14.

In addition to logistical efficiency and costs, fuel consumption plays an important role in assessing the environmental impact and sustainability of each transport strategy

(Figure 15). While vans offer flexibility and minimize storage requirements, they consume the most fuel (approximately 253 L) due to the large number of trips required to meet delivery and shipping requirements. This makes van transportation the least sustainable option in terms of fuel efficiency and carbon emissions, especially for long production cycles.

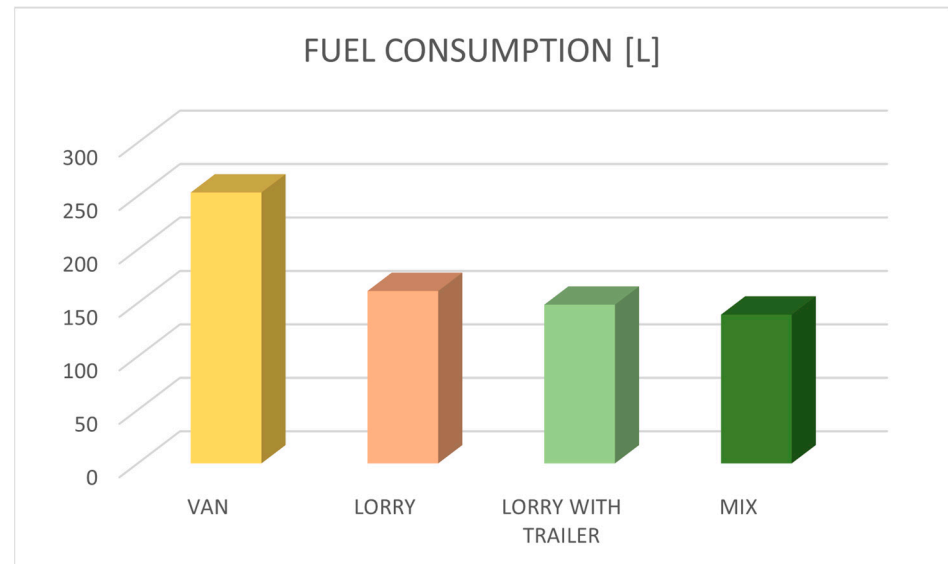


Figure 15. Cost comparison between different types of transport.

Table 8 shows the results of the study. In terms of fuel consumption, TEST_4 proved to be the most sustainable option, as it consumes 82% less fuel due to its high capacity and low number of transport operations.

Table 8. Comparison of overall costs and fuel consumption for all 4 tests.

	TEST_1	TEST_2	TEST_3	TEST_4
Cost of transportation [EUR]	2808	1932	1680	1626
Pallet space rental cost/delivery [EUR]	0	390.6	1398.6	1474.2
Pallet space rental cost/shipment [EUR]	0	693	1751.4	1902.6
Overall costs [EUR]	2808	3015.6	4830	5002.8
Reduction of transportation costs [/]	1.73	1.19	1.03	1
Reduction of overall costs [/]	0.56	0.60	0.97	1
Overall fuel consumption [L]	253.44	161.28	148.8	139.2
Overall fuel consumption efficiency [/]	1.82	1.16	1.07	1

For a market-orientated company, TEST_1 is the best choice in terms of overall costs. The most expensive option is even 78% more expensive in this case. The results show that the implementation of algorithms in a digital twin has a significant impact on the final outcome and can contribute significantly to lowering company costs or reducing greenhouse gas emissions due to lower fuel consumption. The key advantage of the digital twin and the integrated algorithms lies in the extremely fast acquisition of the results from the execution of various what-if scenarios without disrupting the real manufacturing system.

7. Conclusions and Future Work

The research demonstrates the significant potential of integrating digital twin technology with agile supply chain strategies to promote more sustainable manufacturing systems. By providing a dynamic, real-time virtual representation of physical assets and supply chain processes, digital twins enable manufacturers to optimize resource utilization and

improve overall operational resilience. The digital twin framework developed supports informed decision making through predictive analytics and scenario simulations, enabling companies to proactively respond to potential disruptions and implement greener practices.

The tests highlight the importance of considering the trade-offs between transportation efficiency and storage requirements in terms of sustainable supply chain management. While lorries with trailer can reduce transportation costs and fuel consumption, they often increase storage requirements and necessitate external warehousing, which has both economic and environmental implications. In contrast, the use of smaller vehicles such as vans can increase transportation frequency and associated costs, but reduce storage requirements and associated expenses.

The findings underscore the value of digital twins as a powerful tool for manufacturers seeking to balance economic efficiency with sustainability goals. This technology enables data-driven decision making that optimizes resource consumption, reduces emissions and supports the development of more resilient and environmentally conscious supply chains.

Future work will focus on the further development of decision-making algorithms in order to further automate and optimize logistics processes. This will include expanding the capabilities of the digital twin to autonomously manage delivery and dispatch scheduling, minimizing the need for human intervention. The aim is to create a fully self-regulating system that can dynamically adapt to changing conditions, optimize the transport schedule and storage capacities and improve the overall efficiency and sustainability of the supply chain.

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